
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

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Abstract

1 Although reinforcement learning (RL) has attained conspicuous success in aug-
2 menting the reasoning capabilities of language models, the training dynamics that
3 underlie its efficacy in LLMs remain imperfectly understood. In this work, we
4 advance an explanatory account of the RL training process that is grounded jointly
5 in systematic empirical analysis and rigorous theoretical modeling. We first under-
6 take a fine-grained investigation operating simultaneously at the reasoning-pattern
7 and token levels across the trajectory of RL training, from which we establish
8 that, although distinct reasoning patterns retain comparatively stable success rates
9 throughout optimization, RL predominantly reallocates probability mass over a
10 sparse subset of critical tokens and, in so doing, reshapes the distribution over
11 reasoning patterns so as to modulate model performance. Building upon these
12 empirical insights, we construct a theoretical framework that elucidates the train-
13 ing dynamics of RL under two archetypal reward regimes: the verifiable reward
14 (RLVR) and the model’s own internal feedback (RLIF). For RLVR, we characterize
15 the training dynamics through two illustrative cases—one in which models con-
16 verge readily to optimal reasoning strategies, and another in which optimization
17 proves recalcitrant—thereby revealing that the reasoning quality of the base model
18 is decisive for determining convergence behavior. For RLIF, we delineate the mech-
19 anism whereby internal rewards initially confer performance gains yet may, with
20 protracted training, precipitate degradation. Extensive experiments corroborate our
21 findings, thereby advancing both the theoretical understanding and the practical
22 exploitation of RL for language-model enhancement.

23 1 Introduction

24 The recent generation of state-of-the-art reasoning models—among them Gemini2.5 [8], Qwen3 [40],
25 and DeepSeek-R1 [14]—exhibits an exceptional command of intricate logical tasks, spanning both
26 mathematics [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) constitutes the
27 pivotal technique underpinning these advances, having demonstrated a capacity to elevate model
28 competence to a qualitatively new level.

29 The empirical triumphs of RL have, in turn, precipitated a growing body of inquiry into the mecha-
30 nisms that render it effective for LLMs. Drawing on comparisons of pass@k performance, Yue et al.
31 [42] demonstrate that RL post-trained models seldom surpass their base counterparts, an observa-
32 tion that intimates RL may not so much elicit fundamentally novel reasoning patterns as reweight
33 existing ones. Approaching the question through the lens of entropy, Cui et al. [9], Zhang et al. [45]
34 furnish theoretical proof that RL-based methods systematically contract policy entropy, and Cui et al.
35 [9] additionally establish, empirically, a correspondence between model performance and policy
36 entropy. Wang et al. [33], Huan et al. [17] press the analysis further, showing that RL concentrates
37 its optimization upon a sparse subset of critical tokens. Notwithstanding this diversity of efforts to

interrogate RL’s mechanisms, the dynamics governing the RL training process remain only partially elucidated, both empirically and theoretically.

To render the RL training process intelligible, we first undertake a systematic analysis conducted concurrently at the reasoning-pattern and token levels over the course of RL training. Concretely, with the training dynamics as our focal concern, we not only scrutinize the ranking shifts that accrue across training, but also deploy complementary LLM-based and rule-based procedures to extract and taxonomize the reasoning patterns latent in the models’ responses, thereby tracking their attendant success rates and distributions throughout optimization. In contradistinction to prior work [17, 42], our experiments adduce clearer and more compelling evidence that RL predominantly optimizes a sparse subset of critical tokens and, by this means, *reshapes reasoning pattern distributions* so as to govern model performance. We further observe that the intrinsic success rate of any individual pattern remains comparatively invariant. These empirical insights motivate the development of a mathematical framework through which the RL training process may be understood theoretically.

Proceeding from these empirical findings, we further develop a theoretical framework that conceptualizes reasoning as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$, comprising (1) the selection of a reasoning pattern conditioned on the question, i.e., $\pi(r|q)$, and (2) the generation of an answer conditioned on the selected pattern, i.e., $\pi(a|r, q)$. Within this framework, we theoretically dissect the training dynamics of RL in LLMs under two archetypal reward types: the verifiable reward (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4]. For RLVR, we prove that the optimization converges to the reasoning pattern r^* endowed with the highest success rate, a result that accords precisely with our empirical observations. Beyond this, we delineate two qualitatively distinct convergence regimes: models possessing strong initial reasoning quality converge rapidly to optimal patterns, whereas weaker models must first traverse an entanglement stage in which optimization is markedly impeded. For RLIF, we articulate why reinforcement with internal rewards can improve model performance, while simultaneously demonstrating that such optimization may ultimately converge to a state inferior to the base model—a conclusion that coheres with our empirical findings. Supplementary experiments serve to validate the theoretical analysis.

The principal contributions of this paper may be summarized as follows:

- We carry out a systematic analysis, conducted jointly at the reasoning-pattern and token levels, across the RL training process. By conjoining an examination of next-token-prediction ranking shifts with complementary LLM-based and rule-based reasoning-pattern analysis, we furnish clearer and more compelling evidence for understanding RL training dynamics than has been available in prior work [17, 42]. Our experiments establish that RL predominantly optimizes a sparse subset of critical tokens and thereby *reshapes reasoning pattern distributions* so as to modulate model performance, even as the intrinsic success rate of the individual patterns remains comparatively invariant throughout training.
- We formulate a rigorous two-stage mathematical framework that models reasoning as the process $q \rightarrow r \rightarrow a$ (question-reason-answer) and, upon this basis, theoretically analyze the training dynamics of two archetypal RL-based approaches. For RLVR, we prove convergence to the reasoning pattern of highest success rate and distinguish two regimes: rapid convergence for strong base models, as against protracted optimization through an entanglement stage for weaker ones. For RLIF, we supply a theoretical justification for the performance improvements observed at early training stages, while explaining why such methods may ultimately converge to performance inferior to that of the base model.
- We corroborate our theoretical analysis through additional case studies. The proposed framework yields practical insights for understanding and improving RL-based LLM post-training, and in doing so bridges the gap between empirical observation and theoretical understanding of RL training dynamics in LLMs.

2 Related Works

Reinforcement Learning for LLMs Reinforcement learning has proven conspicuously successful in enhancing large language models (LLMs), most notably in aligning model behavior with human preferences [25, 48] and in resolving intricate mathematical and programming tasks [30, 20, 22]. The linchpin of RL is the reward model, which has traditionally been predicated upon human-annotated

91 datasets [26, 25, 2] obtained through extensive training. More recently, however, the prevailing
 92 paradigm has gravitated toward the exploitation of rewards that are at once more readily obtainable
 93 and automatically verifiable, as instantiated in reinforcement learning with verifiable rewards (RLVR)
 94 [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. This success has,
 95 moreover, catalyzed the proliferation of new RL algorithms, among them GRPO [30], DAPO [41],
 96 and VAPO [43].

97 **Theoretical Analyses and Mechanism Interpretation of RL for LLMs.** The efficacy of RL in
 98 LLMs has likewise engendered an extensive literature devoted to theoretical analysis and mechanistic
 99 interpretation. The earliest efforts were largely confined to a high-level treatment of RL, addressing
 100 such questions as how better rewards might be designed [28, 18, 38] and how novel training algorithms
 101 might be devised [37, 11, 21]. As the operative paradigms have shifted toward simpler reward
 102 structures of the RLVR and RLIF variety, researchers have endeavored to apprehend the fundamental
 103 sources of their effectiveness from a plurality of perspectives. Shao et al. [29] revealed that spurious
 104 rewards enhance reasoning by unlocking skills already latent from pretraining, whereas Agarwal et al.
 105 [4] accounted for the observed performance improvements in terms of entropy minimization. Gandhi
 106 et al. [13] identified particular cognitive behaviors, backtracking chief among them, that conduce to
 107 improved reasoning. Further insight was contributed by [46], who demonstrated that RL fine-tuning
 108 amplifies behaviors acquired during pre-training and thereby drives convergence toward dominant
 109 output formats—a phenomenon subsequently analyzed by Wang et al. [34], Cui et al. [10] through
 110 the lens of entropy collapse. In departure from this prior work, our study advances an analysis of RL
 111 training dynamics from the vantage of reasoning pattern selection, underwritten by a mathematical
 112 framework with accompanying theoretical analysis.

113 3 Preliminaries

114 **Reinforcement Learning for LLMs** Let π_θ denote a language model parameterized by θ , which
 115 we take to be the policy subject to optimization. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the
 116 policy π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$, and the optimization objective may accordingly
 117 be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

118 where π_{ref} designates the base reference policy and β is a hyperparameter regulating the KL diver-
 119 gence so as to forestall excessive deviation of π_θ from π_{ref} .

120 The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ admits realization in a variety of forms, whether as a trained reward
 121 model [25] or as a rule-based scoring function [30]. In the present work, we concentrate on two
 122 instances of particular consequence: the verifiable reward (RLVR) [14, 30] and the model’s internal
 123 feedback (RLIF) [47, 4].

124 Under RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ adjudicates directly whether the answer $\mathbf{y}$ coincides with
 125 the correct answer to question $\mathbf{x}$, a typical instantiation being:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

126 In contradistinction to RLVR, RLIF [47, 4] predicates its rewards exclusively upon intrinsic, model-
 127 derived signals. Following Zhao et al. [47], we construe the RLIF reward as the negative average
 128 KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token
 129 distribution

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

130 In this paper, we set ourselves the task of elucidating the training dynamics of both RLIF and RLVR
 131 through conjoined empirical and theoretical analysis.

132 4 Experimental Exploration for RL Training

133 In order to render the RL training process intelligible, we undertake a systematic program of
 134 experiments and analysis spanning RL training under distinct reward regimes. Our investigation

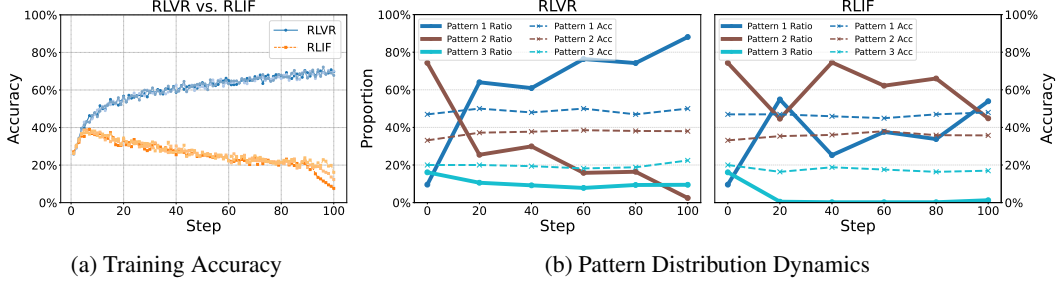


Figure 1: **(a)** The training procedures under RLVR and RLIF, depicting performance on the MATH dataset. The salient finding is that RLVR confers stable gains, whereas RLIF induces an initial increase in performance followed by a subsequent decline; for each training paradigm we conduct multiple rounds of experiments by varying the random seed. **(b)** The reasoning-pattern level comparison. The solid line traces the proportion that a given pattern occupies among all patterns in the responses (left vertical axis), while the dotted line traces the accuracy associated with that pattern (right vertical axis). Over the course of RLVR, the model progressively adopts patterns of higher accuracy and curtails its reliance on those of lower accuracy, whereas RLIF exhibits unstable training dynamics. For ease of inspection, the patterns are sorted and named in descending order of their corresponding accuracy.

135 opens with a high-level overview of the training procedure, from which it emerges that RLVR yields
 136 steady and cumulative improvements, whereas RLIF is liable to instability and may even erode
 137 performance. To anatomize these dynamics, we then conduct a reasoning-pattern level analysis,
 138 which reveals that the success of RLVR is attributable to its propensity to adopt reasoning patterns of
 139 higher success rate, while RLIF fails to specialize in such patterns. Finally, a token-level analysis
 140 interrogates the mechanism underlying these reasoning-pattern dynamics and discloses that they are
 141 driven by shifts in the probability ranks of a surprisingly small fraction of tokens. In comparison
 142 with prior work [17, 42], our experiments furnish clearer and more compelling evidence of these RL
 143 training dynamics, establishing that RL predominantly optimizes a sparse subset of critical tokens
 144 and thereby *reshapes reasoning pattern distributions* so as to modulate model performance.

145 4.1 Training Procedure Overview for RLVR and RLIF

146 We begin by analyzing the complete training procedures of both RLVR and RLIF. To secure a
 147 controlled comparison, we adopt Qwen2.5-3B [39] as the base model and train it on the MATH
 148 dataset [16], holding all other settings fixed and identical across the two algorithms.

149 As Figure 1a illustrates, the two methods trace markedly divergent performance trajectories. RLVR
 150 induces a stable training process, in the course of which model performance improves continuously
 151 and converges gradually. RLIF, by contrast, though it confers gains in the early stages, is prone under
 152 continued training to a subsequent decline in performance, on occasion yielding a model inferior
 153 even to the original base model. These observations furnish the impetus for a more systematic and
 154 fine-grained empirical analysis directed at the underlying RL training dynamics.

155 4.2 Reasoning-pattern Level Analysis

156 To account for the performance disparities observed between RLVR and RLIF, we conduct a fine-
 157 grained analysis at the reasoning-pattern level. In order to define and extract these patterns, we
 158 first assemble responses from the base model and enlist GPT-4o [19] to partition them into distinct
 159 categories on the basis of keywords and logical structure, generating in addition a description for
 160 each pattern. These classifications then serve as the instrument through which we track the reasoning
 161 pattern distribution and its attendant success rates over the entire RL training procedure.

162 Specifically, we characterize the dynamics of the reasoning patterns and their accuracy over the
 163 course of both RLVR and RLIF training. Figure 1b presents our analysis on responses to Number
 164 Theory tasks drawn from the MATH dataset [16], with corroborating results for further models and
 165 datasets deferred to Appendix A. The analysis yields three principal findings:

- RLVR-trained models shift consistently toward the adoption of reasoning patterns of higher success rate, which accounts for the steady improvement in overall accuracy that RLVR effects.
- The reasoning pattern distribution under RLIF displays unstable training dynamics and fails to specialize in the more effective patterns.
- The success rate of any individual reasoning pattern remains stable throughout training under both methods.

This analysis indicates that the performance disparity between RLVR and RLIF is traceable to their divergent reasoning-pattern dynamics. We accordingly pursue further experiments in order to elucidate the mechanisms that drive these changes over the course of RL training.

4.3 Token-level Analysis

Table 1: The proportion of tokens whose probability rank shifts following RL.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

To penetrate the mechanisms underlying the reasoning-pattern dynamics observed during RL training, and taking inspiration from Huan et al. [17], Yue et al. [42], we proceed to a token-level analysis. We first sample responses from the base model for questions drawn from the GSM8K [6], MATH [16], and AIME [7] datasets. Then, holding these responses fixed, we examine the corresponding token ranks in both the base and the RL-enhanced models, where a token’s rank denotes the position its probability occupies among all vocabulary tokens conditioned on the preceding context. Our analysis, summarized in Table 1, reveals that the ranks of individual tokens remain largely invariant, with fewer than 10% of tokens undergoing a rank shift in the majority of cases. This finding intimates that RL selectively perturbs the probabilities of a sparse set of critical tokens while leaving the preponderance of the reasoning process untouched.

Taken together, our experiments furnish clean and compelling evidence concerning the dynamics of RL training: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

These empirical findings not only afford a deeper understanding of RL training dynamics but also supply the groundwork for constructing a mathematical framework in which further theoretical analysis may be pursued.

5 Theoretical Characterization of RL Training Dynamics

Moving beyond empirical observation, we now take a further step toward a theoretical understanding of the RL training process. In this section we first erect a theoretical framework informed by our experimental insights, abstracting the model’s reasoning procedure as a two-step process—the selection of a reasoning pattern followed by the deduction of an answer—in which RL specializes in optimizing the former. On the basis of this framework, we furnish a theoretical account of our observations concerning RL under distinct rewards. For RLVR, we analyze its training dynamics and examine two special cases exhibiting qualitatively distinct optimization behaviors. For RLIF, we explain both why reinforcement with internal rewards can improve model performance and why such methods may ultimately yield models inferior to the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize the reasoning process as follows: given a question q , the model (1) selects a reasoning pattern r from the candidate set $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates, accordingly, a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$. The model selects reasoning pattern r_i with probability $p(r_i|q)$, and each reasoning pattern is endowed with a distinct success rate. Within

213 this framework, the RLIF optimization objective (Eq. 3.3) admits the following reformulation¹:

$$r_\phi(\{\mathbf{r}, \mathbf{a}\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(\mathbf{a}_j|q, \mathbf{r})) \right) \quad (5.1)$$

214 **Policy Parameterization.** Let $\mathcal{V} = \{\mathbf{q}\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and let $\mathcal{Y} =$
 215 $(\mathbf{r}, \mathbf{a})_{\mathbf{r} \in \mathcal{R}, \mathbf{a} \in \mathcal{A}}$ represent the set of output sequences. Given a question $\mathbf{q}$, the language model induces,
 216 autoregressively, a distribution over output sequences $\mathbf{y} \in \mathcal{Y}$:

$$\text{(general policy)} \quad \pi_\theta(\mathbf{y}|\mathbf{x}) = \prod_{l=1}^{|\mathbf{y}|} \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \prod_{l=1}^{|\mathbf{y}|} \text{softmax}(f_\theta(\mathbf{x}, \mathbf{y}_{<l}))_{\mathbf{y}_l}, \quad (5.2)$$

217 where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is a function parameterized by θ , and the model predicts the l -th token
 218 conditioned on the preceding context $\mathbf{y}_{<l}$. The next token is drawn according to the distribution
 219 $\text{softmax}(\mathbf{z})_v := \exp(\mathbf{z}_v) / \sum_{v' \in \mathcal{V}} \exp(\mathbf{z}_{v'})$ for $\mathbf{z} = f_\theta(\mathbf{x}, \mathbf{y}_{<l})$.

220 Owing to the complexity of models encountered in practice, the establishment of optimization
 221 guarantees adequate to an understanding of RL training dynamics has proven notably recalcitrant
 222 [3, 23]. Following prior work [27, 24, 10, 45], we therefore adopt a *tabular policy* parameterization,
 223 which may be regarded as a special case of Eq. 5.2 in which each output is assigned its own trainable
 224 logit for the corresponding last token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$\text{(tabular policy)} \quad \pi_\theta(\mathbf{y}_l|\mathbf{x}, \mathbf{y}_{<l}) = \pi_\theta(\mathbf{y}_l|\mathbf{y}_{l-1}) = \text{softmax}(\theta_{:, \mathbf{y}_{l-1}})_{\mathbf{y}_l}, \quad (5.3)$$

225 where $\theta_{:, \mathbf{y}_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ denotes the column of θ corresponding to $\mathbf{y}_{l-1}$.

226 **Optimization for RL.** For the purposes of our training-dynamics analysis, we work with the tabular
 227 policy of Eq.5.3 under the optimization objective $\phi_{\text{RL}}(\theta)$ of Eq.3.1, and we analyze the policy
 228 gradient in the small-learning-rate limit by means of gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

229 where $\theta(t)$ denotes the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized at $\theta(0) = \theta_{\text{ref}}$.

230 Our experiments disclosed that the success rates of individual reasoning patterns tend to remain stable
 231 throughout training. We ascribe this stability to the model’s architectural constraints: optimizing
 232 the mapping from questions to reasoning patterns is substantially more tractable than optimizing the
 233 path from reasoning patterns to final answers. To render this observation precise, we introduce the
 234 following assumption:

235 **Assumption 5.1.** *The success rate for each reasoning pattern $\mathbf{r}_i \in \mathcal{R}$ to provide the correct answer*
 236 *$\mathbf{r}^*$, defined as $\pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r}_i)$, remains constant during training.*

237 Throughout the remainder of this section, we adopt Assumption 5.1 and write $p^*(\mathbf{r}) = \pi_\theta(\mathbf{a}^*|\mathbf{q}, \mathbf{r})$
 238 for the fixed success rate of a given pattern $\mathbf{r}$. We first derive the optimal policy for the RL objective
 239 (Eq.3.1) under the general autoregressive policy (Eq.5.2), and we subsequently analyze the RLVR
 240 and RLIF training dynamics under the tabular policy (Eq. 5.3).

241 5.2 The Optimal Policy for RL

242 Our empirical results demonstrate that RLVR improves model performance steadily by incentivizing
 243 reasoning patterns of higher success rate, whereas RLIF effects only an unstable improvement. Here
 244 we begin by providing a theoretical characterization of the optimal policy for the RLVR and RLIF
 245 objectives:

246 **Proposition 5.2.** *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive*
 247 *policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

¹For simplicity, we omit the normalizing coefficient $1/|\mathcal{Y}|$ from Eq. 3.3, which may be treated as a constant $1/2$ within our framework.

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 characterizes the optimal solutions for RLVR and RLIF. Presuming that π_{opt} resides within the space parameterized by the general autoregressive policy—a condition invariably satisfied for both LLMs and the tabular policies considered subsequently—the RLVR-optimized model elevates the probability of a reasoning pattern $\mathbf{r}$ precisely when its probability of deducing the correct answer, $p^*(\mathbf{r})$, is sufficiently large that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. For RLIF, by contrast, the formulation of $R_{\text{RLIF}}(\mathbf{r})$ registers only the confidence of the final-answer distribution and thus fails to discriminate between the correct answer $\mathbf{a}^*$ and its incorrect alternatives. In consequence, whereas the optimal policy under RLVR consistently conduces to improved performance, the optimal policy under RLIF affords no guarantee of accuracy exceeding that of the base model.

Moreover, since β is in practice typically small (e.g., 0.001) [26, 5, 44], in the limit $\beta \rightarrow 0$ we obtain:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit the policy collapses to a deterministic strategy that invariably selects the reasoning pattern $\mathbf{r}$ of highest $R(\mathbf{r})$, irrespective of the manner in which the initialized reference model apportions probability among reasoning patterns. Owing to the non-convexity of the optimization landscape, however, the dynamics by which RLVR reliably locates high-reward patterns, and the reasons for which RLIF may exhibit early performance improvements, remain opaque. To dispel this obscurity, we next analyze the training dynamics of both methods under the tabular policy (Eq. 5.3), thereby affording further insight into the RL training process.

5.3 Training Dynamic Analysis for RLVR

Our experiments demonstrate that the improvements conferred by RLVR are heavily contingent upon the capacity of the base model. Here we direct our attention to the probability of the *optimal reasoning pattern*, namely the pattern $\mathbf{r}^*$ possessing the highest success rate for arriving at the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$. Analyzing the gradient flow, we uncover two distinct regimes in the training dynamics of RLVR. In the first, the probability of the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically until it converges to 1. In the second, the model initially traverses an entanglement stage in which a suboptimal reasoning pattern $\mathbf{r}' \neq \mathbf{r}^*$ impedes the optimization of the optimal pattern; only upon emerging from this entanglement stage does the model transition to the dynamics of the first regime and converge to the optimal reasoning pattern $\mathbf{r}^*$.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 treats the case in which the base reference model is sufficiently strong that its overall accuracy exceeds the success rate of every non-optimal reasoning pattern (Eq 5.7). Under this condition, RLVR efficiently steers the model toward the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$, thereby attaining a high overall accuracy that approaches the success rate of the optimal reasoning pattern.

Practical experience attests, however, that RLVR optimization can in certain scenarios prove challenging [44, 36], a difficulty that arises characteristically when the base reference model is less capable. We therefore turn to the second regime, in which the model first traverses an entanglement stage wherein the *suboptimal reasoning pattern of second-highest success rate*, defined as $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, retards the optimization of the optimal reasoning pattern:

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

Theorem 5.4 concerns a special regime in which only the success rates of the optimal and suboptimal reasoning patterns exceed the overall accuracy. We confine our attention to this case for the sake of analytical tractability, and we hold that it may be generalized—albeit at the cost of a more intricate analysis—to settings in which *at least* two reasoning patterns are permitted success rates above the average.

The critical insight afforded by Theorem 5.4 is, further, that RLVR may require T_0 time steps before the overall success rate comes to exceed that of the suboptimal reasoning pattern—that is, before the model attains the regime of Theorem 5.3, in which the selection probability of the optimal reasoning pattern is sufficiently large. Whereas Theorem 5.3 establishes that the convergence time T_1 is polynomial in $1/\epsilon$, the time T_0 of Theorem 5.4 may grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$ —the ratio of the aggregate success rate through $\mathcal{R}/\{\mathbf{r}'\}$ to the success rate through $\mathbf{r}^*$. Manifestly, when the base model apportions only a vanishingly small selection probability to the optimal reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ may become very large, entailing a prohibitively large T_0 . As a consequence, an extraordinarily protracted training period may be required before the model selects the optimal reasoning pattern with appreciable probability—a phenomenon we term the entanglement stage.

We provide additional experiments and case studies further illustrating these two regimes in Sec 6.

5.4 Theoretical Explanation for RLIF

In Proposition 5.2 we established that RLIF, attending as it does only to the confidence of the final-answer distribution, fails to discriminate between the correct answer $\mathbf{a}^*$ and its incorrect alternatives. This intimates that RLIF may in due course precipitate a degradation of performance relative to the base model, a prospect that accords well with our empirical observations. Prior studies have nonetheless demonstrated that RLIF can improve model performance in the absence of external rewards Agarwal et al. [4], Zhao et al. [47]. Here we furnish a theoretical explanation of why RLIF is able to improve the performance of a well-trained LLM at the initial stage of training.

We first consider a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|\mathbf{q}, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|\mathbf{q}) = \mathbf{a}^*$.*

Since majority voting [35] has proven an effective means of improving model performance—and since well-trained LLMs consequently tend to assign the correct answer $\mathbf{a}^*$ a higher probability than any incorrect answer—Assumption 5.5 is likely to obtain for modern LLMs. Under this constraint, we examine the case in which $|\mathcal{A}| = 2$, the success rate of each reasoning path follows a uniform distribution, and the reasoning-pattern selection is of high entropy². Within this setting, we analyze the overall-accuracy dynamics at $t = 0$, obtaining the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

²Prior work [33] reveals that RL primarily optimizes high-entropy tokens. Here we consider the case of maximal entropy: $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

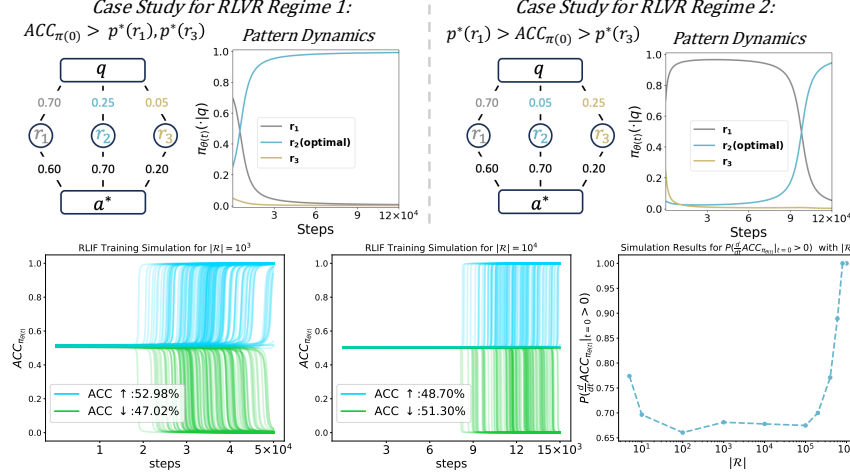


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom). The upper panels exhibit the two distinct training regimes of RLVR: rapid convergence (left) and the entanglement phase (right). The bottom panels report, for RLIF and across varying $|\mathcal{R}|$, the probabilities of the model’s convergence to distinct terminal states (left) together with the probability of an initial performance gain (right).

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt}ACC_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The first result of Theorem 5.6 shows that overall accuracy increases at initialization, thereby supplying a theoretical explanation for the early performance gains of RLIF. As training progresses, however, Proposition 5.2 implies that when $\beta = 0$ the policy converges deterministically to the reasoning pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second result of Theorem 5.6 reveals that, with probability 0.5, this maximizing pattern coincides with the one that *minimizes* the success rate: $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. Stated otherwise, there obtains a 50% chance that the model converges to the least accurate reasoning path. Consequently, although RLIF improves model performance at the outset (the derivative of accuracy being positive), continued training may drive the model toward states that privilege reasoning paths of very low accuracy, culminating in performance inferior to that of the base model.

6 Case Studies and Numerical Simulations

To further validate and interpret our theoretical findings, we adduce case studies and training simulations that illustrate our theorems. For RLVR, we exhibit the two distinct training regimes treated in Theorems 5.3 and 5.4, as depicted in Fig. 2 (upper). For RLIF, we conduct a battery of training simulations in which the success rate of each reasoning path follows a uniform distribution, thereby corroborating our analysis of RLIF’s behavior both during the initial training steps and at convergence, as shown in Fig. 2 (bottom).

6.1 Case Studies for RLVR

In Section 5.3 we distinguished two regimes in the training dynamics of RLVR. In the first (Eq 5.7), the probability of selecting the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ increases monotonically until convergence. In the second (Eq 5.8), the model undergoes an initial entanglement phase and demands a substantially longer training period before it transitions to the dynamics of the first regime and converges to the optimal reasoning pattern $\mathbf{r}^*$. We consider each regime in turn:

- As Fig. 2 (upper left) demonstrates, when the reference model satisfies the conditions of Eq 5.7—so that the overall accuracy exceeds the success rates of all non-optimal reasoning patterns—the model converges rapidly to the optimal reasoning pattern.
- In the second case, corresponding to Theorem 5.4, in which a suboptimal reasoning pattern $\mathbf{r}'$ attains a success rate exceeding the initial overall accuracy, the model first undergoes an entanglement phase, and a transition period T_0 (defined in Theorem 5.4) must elapse before it enters the rapid

optimization phase characteristic of the first case. Fig. 2 (upper right) portrays a scenario in which $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9) is large ($\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$). In such instances the transition time T_0 , which scales as $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, becomes prohibitively long and substantially delays convergence to the optimal reasoning pattern. The pattern dynamics of Fig. 2 manifest precisely such an entanglement stage, in close agreement with our theoretical result.

6.2 Numerical Simulations For RLIF

In Section 5.4 we demonstrated that, under Assumption 5.5 and the base-model distribution specified in Theorem 5.6, the accuracy derivative at $t = 0$ is positive with probability 1 as $|\mathcal{R}| \rightarrow \infty$, while there subsists a 50% probability of convergence to the least accurate reasoning path. Our simulations bear out both findings. Fig. 2 (bottom left) presents RLIF training simulations across varying $|\mathcal{R}|$, the results of which confirm that the probability of convergence to an improved state ($ACC \uparrow$) approximately equals that of convergence to a degraded state ($ACC \downarrow$), thereby validating the second result of Theorem 5.6. Our examination of the initial-step performance, displayed in Fig. 2 (bottom right), further demonstrates that for large $|\mathcal{R}|$ the probability of an initial performance improvement approaches 1, in support of the first result of Theorem 5.6.

7 Conclusions and Limitations

This work interrogates the dynamics of reinforcement learning in LLMs through the conjunction of empirical investigation and theoretical framework. We develop mathematical analyses for two representative reward mechanisms—RLVR and RLIF—and corroborate our findings through case studies and simulations. Our analysis is nonetheless attended by certain limitations that merit further inquiry: the interpretability of the reasoning patterns identified by LLMs stands in need of additional validation; our theoretical framework could be extended to accommodate the more intricate reasoning scenarios encountered in practice; and the present analysis could be generalized beyond the specific assumptions imposed upon the base model.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

In Section 4.2 we sampled Number Theory tasks from MATH in order to analyze the dynamics of the reasoning patterns. Here we additionally sample Geometry tasks for the same dynamic analysis (Figure 3). Over the course of RLVR, the model progressively adopts the reasoning patterns of higher accuracy, whereas the pattern selection under RLIF proves unstable. This outcome is fully consistent with the empirical and theoretical results advanced in the main paper.

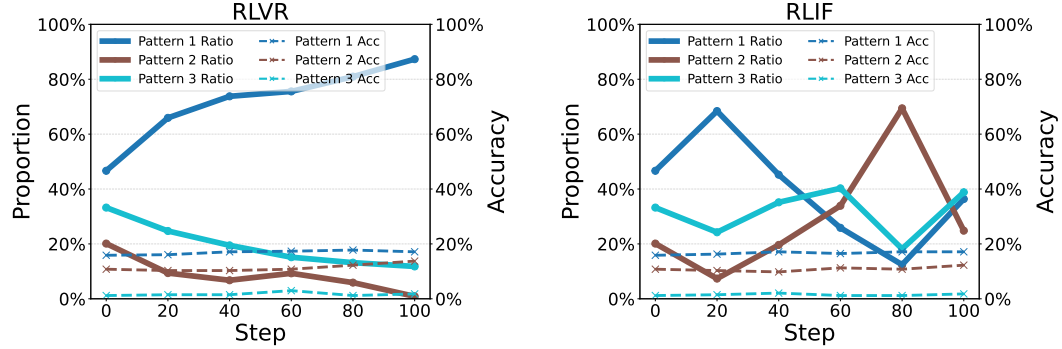


Figure 3: Dynamics of the pattern distribution and its corresponding accuracy for RL-enhanced models on Geometry tasks. Under RLVR, the model gravitates toward the pattern of highest accuracy (Pattern 1), whereas RLIF remains unstable.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

In order to ascertain whether our findings extend to different models, we conduct supplementary experiments that put the experimental insights and theoretical conclusions to the test (Figure 4). For these experiments we take Qwen-2.5-7B-Instruct [39] as the base model and set it against Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced through RLVR for mathematical reasoning. Our evaluation traverses a diversity of mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—drawing challenging problems from the MATH dataset [16], and we extend the analysis further to encompass the more complex problems of AMC23 [1]. The results uniformly reinforce our earlier findings: RLVR enhancement elevates the frequency of high-accuracy reasoning patterns while attenuating that of the less effective ones. Detailed task-specific analyses are deferred to Appendix B.3.

Across the reasoning-pattern analysis conducted over these varied tasks, we observe that the patterns of highest success rate consistently grow more prevalent following RLVR enhancement, a regularity that reinforces our findings from Section 4 and Section 5. Notably, in tasks such as Algebra and AMC23-19 in Figure 4, the success rates of the individual reasoning patterns remain consistent, an observation that not only comports with our earlier conclusions but also furnishes empirical warrant for Assumption 5.1 in our theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To subject our theory to still more stringent scrutiny, we conduct experiments on a larger model, QwQ-32B [32], which is built upon Qwen-2.5-32B-Instruct [39] and whose reasoning ability is substantially augmented through RLVR. We examine the shift in reasoning patterns between these two models on four tasks: the number theory and geometry tasks of the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

It is apparent that the preponderance of QwQ-32B’s reasoning patterns correspond to the most accurate patterns of Qwen-2.5-32B, in close agreement with our theoretical results. We refrain from reporting the accuracy of QwQ because our reliance on the models’ APIs for testing precludes the extraction of the CoT data that would be required to have the base model continue generating answers; the accuracy of the reasoning patterns (as distinct from the accuracy of the model itself) is thereby

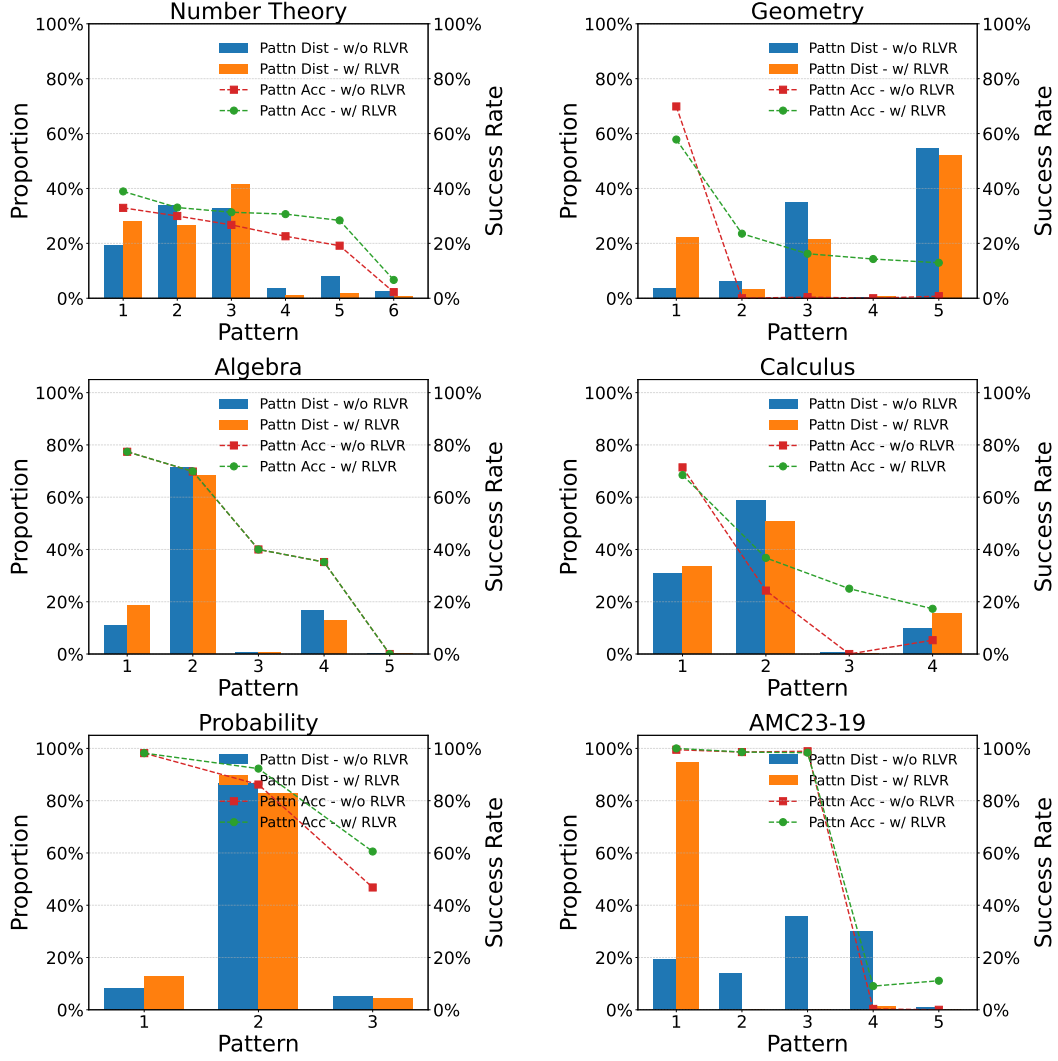


Figure 4: Evaluation results for the reasoning patterns and their corresponding success rates, for models with and without RLVR enhancement, across a range of additional tasks; these results accord closely with our experimental insights and theoretical conclusions. The bar (Pattern Dist) denotes the proportion that a given pattern occupies among all patterns, while the dot (Pattern Acc) denotes the accuracy, or success rate, associated with that pattern.

rendered unobtainable. For further details, we refer the reader to “Accuracy Analysis” in Appendix B.2.

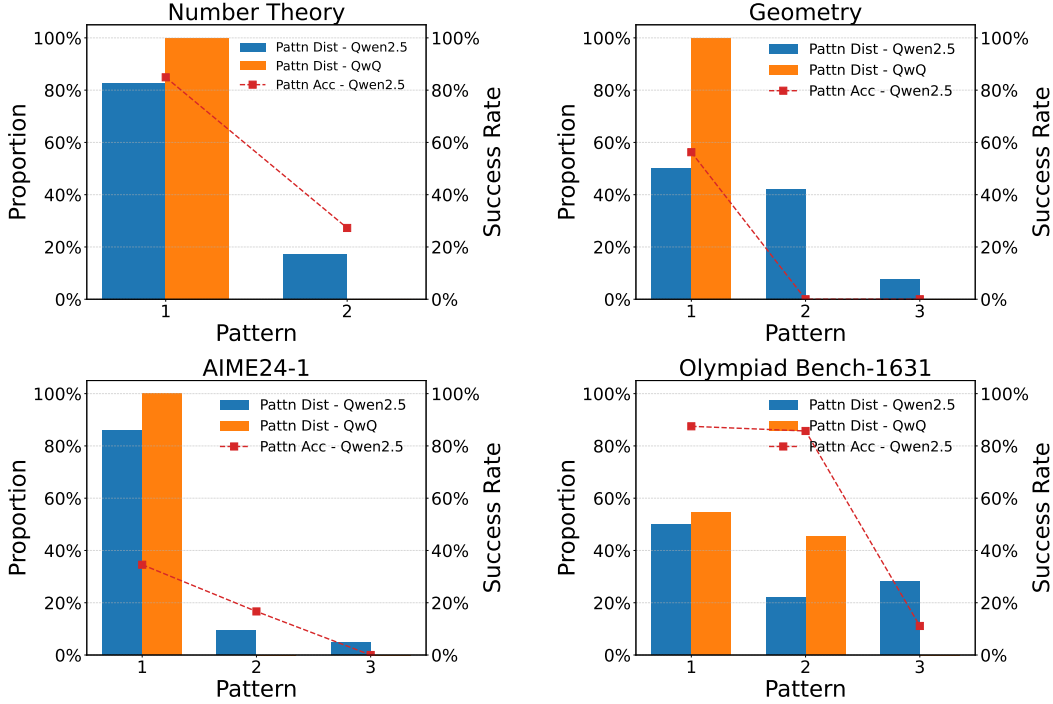


Figure 5: Evaluation results on the larger models Qwen-2.5-32B and QwQ-32B. The reasoning patterns of QwQ converge to the most accurate pattern of Qwen-2.5, thereby attesting to the applicability of our theory at larger scales.

B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: We first adopt Qwen2.5-3B as the base model and train it on the MATH dataset under both RLVR and RLIF. To this end we employ the verl [31] framework, ensuring that all parameter settings coincide with those of the MATH-dataset example furnished by the framework; for RLIF, the reward-calculation procedure alone is altered. By varying the random seed, we obtain three distinct training curves for each method.

Token-level Analysis: This experiment employs Qwen2.5-3B together with the RL-enhanced models described above. For GSM8K and MATH we randomly sample 400 questions apiece from their respective test sets, while for AIME24 we use the dataset in its entirety. We first sample from the base model the corresponding answer to each question; we then concatenate this answer to the question itself and supply the result to the RL-enhanced model as a new prompt. Availing ourselves of the interface provided by the OpenAI library, we compute the probability of each token in the prompt together with its rank among the probabilities of all admissible tokens at the current position. By adjudging whether the rank of each token has shifted before and after RL, we arrive at the proportion of tokens whose rank has changed.

Reasoning-pattern Level Analysis: For this purpose we conduct two species of experiment. We first study the dynamics of the pattern distribution over the course of training, using the previously trained models—namely Qwen2.5-3B and its RL-enhanced counterpart—and evaluating at intervals of 20 steps. To gauge the applicability of our conclusions to other base models and to larger models, we additionally test the open-source models Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B alongside their respective base models; for these, we are able to examine only the change in the pattern

distribution of the model outputs before and after RL, and not the dynamics of the distribution over the training process itself.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: We first randomly sample questions of differing task type (Geometry, Algebra, and so forth) and differing difficulty level (Level 4, Level 5) from datasets including MATH. For each question, the model samples 1024 answers in the pattern-distribution-dynamics task, and—depending on the task category—between 64 and 1024 answers in the task that considers only the distributional change before and after RL. These responses subsequently serve as the material for pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of the base model’s responses and enlist the GPT-4o [19] API to distill pattern categories from these samples. The full prompt employed for reasoning-pattern extraction is provided in Appendix D.1. To safeguard the fidelity of the extraction, we fix the temperature of GPT-4o’s API at 0.

Responses Classification: We again enlist GPT-4o, at temperature 0, to classify the full set of responses according to the pattern categories thus identified. The full prompt employed for response classification is provided in Appendix D.2.

Accuracy Analysis: So as to isolate the impact of the pattern distribution upon accuracy, we construct new prompts by concatenating the original prompt with partial responses from the base model, on the supposition that these partial responses adequately instantiate specific reasoning patterns. This device holds the reasoning process fixed and thereby facilitates our determination of whether the change in model accuracy is attributable solely to the change in the distribution over reasoning patterns, rather than to any change in the reasoning process itself. Concretely, we excise from the base model’s response the sentence containing the final answer, append the sentence immediately preceding it to the end of the question, and present the result as a new prompt to the RL-enhanced model, which then continues to predict the answer. The accuracy of this predicted answer we take as the accuracy of the reasoning pattern.

B.3 Details for Each Task

In what follows we set out the experimental procedures for each task in detail, encompassing the task descriptions together with representative examples of the reasoning patterns (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: This task confronts the models with problems of coin distribution across multiple bags. The bags initially contain equal numbers of coins; upon the receipt of additional coins and their equal redistribution, the total is required to exceed a specified value while equality of distribution is preserved, and the models must determine the minimum initial count of coins per bag. We derived the task from a level-5 MATH problem (id: test/number_theory/1055) [16], generating 32 variants—by adjusting such parameters as the bag count and the coin totals—to constitute our evaluation dataset. We adopt one-shot prompting with a simple and unrelated example so as to guide answer formatting via “boxed” notation without prejudicing the reasoning approaches. For evaluation, we first select 20 questions from the evaluation dataset with 4 responses each for reasoning-pattern extraction, sampling 64 responses per question. We thereafter categorize all responses by the extracted reasoning patterns, which allows us to compare the change in the distribution of the models’ reasoning patterns before and after RL.

Geometry: For the geometry task, the model is furnished with the lengths of two sides of a scalene triangle and asked how many distinct integer centimeter lengths the third side may assume. This task derives from a level-4 geometry problem in MATH (id: test/geometry/1046). Our evaluation adheres to the same pipeline as in Number Theory: we construct a synthetic dataset of 32 questions, sample 20 questions and select 4 responses each from the base model (Qwen2.5-7b-Instruct) for reasoning-pattern extraction, and then compare the responses of the model with and without RLVR enhancement over all questions in the evaluation dataset, with 64 responses each, for pattern analysis.

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative examples of reasoning patterns across the various tasks.

Algebra: The algebra task we employ is a can-recycling problem. We begin with a certain number of old cans, and it is stipulated that n old cans may be recycled into a single new can; the question is how many cans may ultimately be produced. The difficulty resides in the fact that at each step of the iterative calculation there may remain surplus cans that do not divide evenly, and these may eventually be combined for further recycling. The template for this task derives from a level-4 algebra problem in MATH (id: test/algebra/2768). For all the supplementary tasks that follow, beginning with this one, we adopt the same settings as in the preceding tasks, including the dataset size, the number of samples, and so forth.

Calculus: Our calculus task is simple and direct. We present a trigonometric equation in a single variable together with the domain of that variable, and ask the model to determine the number of solutions to the equation. Though elementary, the problem serves to exhibit the model’s basic competence on calculus problems. Its prototype is a level-4 pre-calculus problem in MATH (id: test/precalculus/1140).

Counting and Probability: We further study the model’s performance on the counting-and-probability task. Given a collection of white and black balls, from which two balls are drawn at random, we specify the probability that one of the two drawn balls is black and the other white. We also provide the number of balls of one color and require the model to calculate the minimum number of balls of the other. This probability-theoretic task derives from a level-4 counting-and-probability problem in MATH (id: test/counting_and_probability/79).

673 **AMC23 Question:** Here we select question 19 of AMC23 [1]; for reasoning-pattern extraction we
 674 randomly sample 48 responses, and we subsequently sample 1024 responses for the reasoning-pattern
 675 analysis.

676 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

677 We conduct experiments on four tasks. Owing to the constraints attendant upon API-based testing, we
 678 randomly select a single question per task as input, sample 64 responses from each of the two models,
 679 and perform reasoning-pattern extraction and classification directly from these answers. Specifically,
 680 for the two MATH-dataset tasks we select question 1055 from the number theory task and question
 681 1046 from the geometry task, while for the remaining two we take problem 1 from AIME24 and
 682 question 1631 from OlympiadBench. Examples of the patterns are exhibited in Table 2.

683 C Deferred Proofs

684 In this appendix we furnish the proofs of our principal theoretical results: Proposition 5.2, Theo-
 685 rems 5.3, Theorems 5.4 and Theorems 5.6.

686 C.1 Proof of Proposition 5.2

687 *Proof of Proposition 5.2.* In this proof, we utilize the proof techniques in [26], recall that the opti-
 688 mization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

689 Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample
 690 $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$
 691 can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{r}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

692 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

693 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the
 694 partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

695 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

696 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

697 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
 698 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \log|\mathcal{A}| \right)}_c, \end{aligned}$$

where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|q) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|q)} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|q)} \end{aligned}$$

So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r}))$. This concludes the proof of the theorem. $\square$

C.2 Dynamics of Tabular Policy

Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the gradient dynamics of θ can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|q)} [r_{\phi}^{KL}(\mathbf{r}|q)] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|q) \nabla \pi_{\theta(t)}(\mathbf{r}|q)], \end{aligned}$$

Where $\pi_{\theta(t)}(\mathbf{r}|q) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to q , and $r_{\phi}(\mathbf{r}|q)^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, q}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|q) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|q) (-\pi_{\theta(t)}(\mathbf{r}_j|q) \cdot \pi_{\theta(t)}(\mathbf{r}_i|q))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|q) \cdot \pi_{\theta(t)}(\mathbf{r}_i|q) \end{aligned}$$

where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|q) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|q)$, we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|q)}{\partial \theta_{\mathbf{r}_j, q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, q}(t)}{\partial t}$$

C.3 Proof of Theorem 5.3

To prove Theorem 5.3, we first establish that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4); we then derive a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|q)$ (Eq C.5); and finally we obtain the bound on t asserted in Theorem 5.3.

714 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

715 For $\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}), \end{aligned}$$

716 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \end{aligned} \quad (\text{C.4})$$

717 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
718 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
719 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \end{aligned} \quad (\text{C.5})$$

720 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
721 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

722 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

723 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

724 This concludes the proof of the theorem. $\square$

725 **C.4 Proof of Theorem 5.4**

726 To prove Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6); we then analyze
 727 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7) and demonstrate that, for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

728 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 729 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}')\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

730 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

731 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt}\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

732 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 733 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

734 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

735 Then we have:

$$\frac{d}{dt}\rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

736 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

737 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

738 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

739 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

740 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

741 This concludes the proof of the theorem. $\square$

742 C.5 Proof of Theorem 5.6

743 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
744 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

745 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 746 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

747 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 748 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 749 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 750 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

751 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 752 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 753 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 754 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3} \delta > 0$$

755 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 756 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

757 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 758 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

759 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

760 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 761 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 762 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 763 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 764 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 765 $1 - 6\delta$.
- 766 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 767 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

768 This concludes the proof of the second statement.

769

□

770 D Full Prompts

771 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

772

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

773

774 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

775

776 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

777

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

778

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

779

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

780

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

781

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

782

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

783

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

784

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

785

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a s -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

786

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

787